NAME

mbgetphotocorrection − Generate a 3D table of image intensity correction values from a set of navigated stereo or mono seafloor photographs.

VERSION

Version 5.0

SYNOPSIS

mbgetphotocorrection [**−−verbose −−help −−threads=***nthreads* **−−input=***imagelist* **−−output=***file* | **−−correction−file=***file* **−−correction−x−dimension=***value* **−−correction−y−dimension=***value* **−−correction−z−dimension=***value* **−−correction−z−minmax=***value/value* **−−fov−fudgefactor=***factor* **−−trim=***trim_pixels* **−−reference−gain=***gain* **−−reference−exposure=***exposure* **−−platform−file=***platform.plf* **−−camera−sensor=***id* **−−nav−sensor=***id* **−−sensordepth−sensor=***id* **−−heading−sensor=***id* **−−altitude−sensor=***id* **−−attitude−sensor=***id* **−−use−left−camera** | **−−use−right−camera** | **−−use−both−cameras −−calibration−file=***stereocalibration.yaml* **−−navigation−file=***file* **−−tide−file=***file* **−−image−quality−file=***file* **−−image−quality−threshold=***value* **−−image−quality−filter−length=***value* **−−topography−grid=***file*]

DESCRIPTION

Mbgetphotocorrection generates a 3D table of image intensity correction values, indexed by lateral pixel position (x, y with respect to the camera image) and by standoff distance (z, the range from the camera to the seafloor along the line of sight). This table is intended for later use by **mbphotomosaic** to flatten illumination variation across a photomosaic built from many overlapping downlooking photographs.

The program reads a sequence of single or stereo pair images and associated per-image navigation, attitude, and camera settings from an MB-System imagelist file (in the format written by **mbm_makeimagelist** or similar tools). For each image, the camera position and orientation are combined with an optional platform offset model and an optional bathymetric topography grid (or, absent a topography grid, an assumed flat seafloor at a nominal standoff) to project every pixel onto the seafloor and calculate that pixel's standoff distance. Each image is optionally undistorted using a stereo camera calibration model before being processed. Pixels are excluded from the correction table if their line of sight is too vertical (more than 80 degrees from vertical, since this program assumes a 2D downlooking photomosaic geometry) or if the standoff falls outside the requested z range. Purely black pixels, and pixels adjacent to purely black pixels, are also excluded unless an explicit pixel trim margin is set with **−−trim**.

Prior to binning, each pixel's YCrCb intensity is corrected for the camera's recorded gain and exposure settings relative to user-specified reference gain and exposure values, so that images captured under different camera settings contribute consistently to the same correction table.

The 3D correction table (separate tables are maintained for each of the two cameras of a stereo pair) accumulates the mean Y, Cr, and Cb values in each x/y/z bin. After all images have been processed, any empty bins are filled in by two iterative passes: first averaging over neighboring x/y bins at the same standoff, then averaging over neighboring standoff bins at the same x/y location. The completed correction tables are written to an OpenCV YAML file using **cv::FileStorage**, together with the correction table dimensions and bounds and the reference gain and exposure values used. This file can be supplied to **mbphotomosaic** with its own **−−correction−file** option to correct pixel intensities using this table.

Mbgetphotocorrection can process images using multiple threads (see **−−threads**). Any parameter change statements embedded in the imagelist file are honored, causing all in-flight threads to complete before the new parameters take effect.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−help

This "help" flag causes the program to print out a description of its operation and then exit immediately.

−−verbose

The **−−verbose** option causes **mbgetphotocorrection** to print out status and debug messages. Using it more than once increases the verbosity level.

−−threads=*nthreads*

Sets the number of processing threads used to work through the image list. The requested number of threads is limited to the number of hardware threads available and to the MB-System maximum thread count. The default is 1 thread.

−−input=*imagelist*

Sets the filename of the input MB-System imagelist file listing the images (single or stereo pairs) to be processed, along with their navigation, attitude, and camera settings. The default input filename is *imagelist.mb−1*.

−−output=*file*

−−correction−file=*file*

These two equivalent options set the filename of the output OpenCV YAML file that will contain the 3D image correction tables. The default output filename is *imagelist_cameracorrection.yml*.

−−correction−x−dimension=*value*

Sets the number of bins along the image x (column) dimension of the correction table. The default is 11.

−−correction−y−dimension=*value*

Sets the number of bins along the image y (row) dimension of the correction table. The default is 11.

−−correction−z−dimension=*value*

Sets the number of bins along the standoff (z) dimension of the correction table. The default is 41.

−−correction−z−minmax=*value*/*value*

Sets the minimum and maximum standoff distance in meters spanned by the correction table's z dimension; pixels with a calculated standoff outside this range are not used. The default range is 1.0 to 9.0 meters.

−−fov−fudgefactor=*factor*

Sets a multiplicative fudge factor applied to the camera field of view angles when calculating the reference depth used to derive per-pixel takeoff angles. The default value is 1.0 (no adjustment).

−−trim=*trim_pixels*

Sets the number of pixels to ignore around the margins of each undistorted image. If this option is not used (the default, 0 pixels), the program instead automatically ignores any purely black pixel and any pixel adjacent to a purely black pixel, which is intended to exclude the black borders that can appear around the edges of an undistorted image.

−−reference−gain=*gain*

Sets the reference camera gain value, in dB, to which all image intensities are normalized before being added to the correction table. The default is 14.0.

−−reference−exposure=*exposure*
Sets the reference camera exposure time, in microseconds, to which all image intensities are normalized before being added to the correction table. The default is 8000.0.

−−platform−file=*platform.plf*
Sets the filename of an MB-System platform definition file describing the geometric offsets between the vehicle's navigation, attitude, and camera sensors. The platform model is (re)loaded the first time an image is processed after this option takes effect.

−−camera−sensor=*camera_sensor_id*
Sets the platform sensor id number of the stereo camera used to look up the camera's position and orientation offsets within the platform model.

−−nav−sensor=*nav_sensor_id*
Overrides the platform model's navigation (position) source sensor id number.

−−sensordepth−sensor=*sensordepth_sensor_id*
Overrides the platform model's pressure/depth source sensor id number.

−−heading−sensor=*heading_sensor_id*
Overrides the platform model's heading source sensor id number.

−−altitude−sensor=*altitude_sensor_id*
Sets the platform sensor id number used as the altitude source. This value is parsed but is not applied to the platform model by the current implementation.

−−attitude−sensor=*attitude_sensor_id*
Overrides the platform model's roll/pitch and heave source sensor id number.

−−use−left−camera
Process only the left image of each stereo pair (or single images acquired by the left camera).

−−use−right−camera
Process only the right image of each stereo pair (or single images acquired by the right camera).

−−use−both−cameras
Process both the left and right images of each stereo pair. This is the default camera usage mode.

−−calibration−file=*stereocalibration.yaml*
Sets the filename of an OpenCV YAML file containing the intrinsic and extrinsic stereo camera calibration parameters (camera matrices, distortion coefficients, and the rotation/translation between the two cameras) used to undistort images that are not already rectified.

−−navigation−file=*file*
Sets the filename of an ASCII navigation file used in place of, or to supplement, navigation carried in the imagelist file.

−−tide−file=*file*
Sets the filename of an ASCII tide file (time and tide height pairs) used to apply a tidal correction to the water depth used in the photogrammetric projection.

−−image−quality−file=*file*
Sets the filename of an ASCII file of time-stamped image quality values used to filter out low quality images.

−−image−quality−threshold=*value*
Sets the minimum image quality value (in the range 0 to 1) required for an image to be used in the correction table.

−−image−quality−filter−length=*value*
Sets a time filter length, in seconds, applied when interpolating image quality values from the image quality file.

−−topography−grid=*file*

Sets the filename of a bathymetric grid (in a format read by **mbgrid**/GMT) used to intersect each pixel's line of sight with the actual seafloor topography rather than an assumed flat seafloor at a nominal standoff. Specifying this option enables use of the topography grid.

SEE ALSO

mbsystem(1), **mbphotomosaic**(1), **mbphotogrammetry**(1), **mbimagelist**(1), **mbm_makeimagelist**(1), **mbgrid**(1)

BUGS

Maybe. Maybe not.